

Bayesian Classification of a Two-Dimensional Feature Vector

Given

$$\mu_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad \mu_2 = \begin{bmatrix} 3 \\ 3 \end{bmatrix}$$
$$\Sigma_1 = \begin{bmatrix} 1.1 & 0.3 \\ 0.3 & 1.9 \end{bmatrix}, \quad \Sigma_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$
$$x = \begin{bmatrix} 1.0 \\ 2.2 \end{bmatrix}$$

Assuming equal class priors,

$$P(\omega_1) = P(\omega_2)$$

Bayesian Decision Rule

For multivariate Gaussian distributions,

$$g_i(x) = -\frac{1}{2}(x - \mu_i)^T \Sigma_i^{-1} (x - \mu_i) - \frac{1}{2} \ln |\Sigma_i|$$

Since the priors are equal, the prior term is ignored.

Class ω_1

$$|\Sigma_1| = (1.1)(1.9) - (0.3)^2 = 2.0$$

$$\Sigma_1^{-1} = \frac{1}{2.0} \begin{bmatrix} 1.9 & -0.3 \\ -0.3 & 1.1 \end{bmatrix} = \begin{bmatrix} 0.95 & -0.15 \\ -0.15 & 0.55 \end{bmatrix}$$

$$x - \mu_1 = \begin{bmatrix} 1.0 \\ 2.2 \end{bmatrix}$$

$$(x - \mu_1)^T \Sigma_1^{-1} (x - \mu_1) = 3.03$$

$$g_1(x) = -\frac{1}{2}(3.03) - \frac{1}{2} \ln(2.0) = -1.862$$

Class ω_2

$$|\Sigma_2| = 1$$

$$\Sigma_2^{-1} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$x - \mu_2 = \begin{bmatrix} -2.0 \\ -0.8 \end{bmatrix}$$

$$(x - \mu_2)^T \Sigma_2^{-1} (x - \mu_2) = (-2.0)^2 + (-0.8)^2 = 4.64$$

$$g_2(x) = -\frac{1}{2}(4.64) - \frac{1}{2} \ln(1) = -2.32$$

Decision

$$g_1(x) > g_2(x)$$

The vector $[1.0, 2.2]^T$ is classified as class ω_1